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Woohwangcheongsimwon Prevents High-Fat Diet-Induced Memory Deficits and Induces SIRT1 in Mice

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ABSTRACT Woohwangcheongsimwon (WHC) is a mixture of herbal medicines that is widely prescribed in Korean traditional medicine. SIRT1 is known for its regulatory roles in energy metabolism, oxidative stress, and circadian rhythms. This study was designed to determine whether WHC can increase and mimic the biological reactions of SIRT1 activation. Ten-month-old male mice were divided into four groups: nontreated normal diet (ND), nontreated high-fat diet (HFD), WHC-treated ND, and WHC-treated HFD. Body weight and cognitive functions were evaluated after treatment. The hippocampal expressions of SIRT1 and PGC-1α were also measured. The components of WHC were identified by liquid chromatography. High-fat diet-fed mice gained more weight and demonstrated greater deficits in short-term and long-term cognitive functions. WHC suppressed the deleterious effects of a HFD on weight gain and cognitive decline, but showed no prominent effects on animals fed NDs. The herbal treatment also increased the expression of SIRT1 and PGC-1α in the hippocampus. Despite the induction of hippocampal SIRT1 expression by WHC, resveratrol was not present among the natural compounds identified. This expression might have contributed to the suppression of high-fat diet-induced memory deficits in mice treated with the herbal mixture.

KEYWORDS: • ***high-fat diet*** • ***memory*** • ***natural compounds*** • ***resveratrol*** • ***SIRT1*** • ***Woohwangcheongsimwon***

INTRODUCTION

Metabolic syndrome is a cluster of abnormal metabolic conditions that increase the risk of cognitive impairments and vascular dementia; various prevention trials have been performed to modify the associated metabolic risk factors.[1] Experiments on animals have shown that feeding high-fat diet (HFD) caused cognitive impairments related to cerebrovascular dysfunction by abnormal neurovascular couping and oxidative stress.[2–4] Additionally, deposits of amyloid protein are accelerated by long-term feeding of HFDs to Alzheimer's disease mice, whereas various studies have revealed that cognitive impairment can be prevented through caloric restriction in these animals.[5]

Multiple important links between the sirtuin family of proteins and energy metabolism and inflammation have been suggested. SIRT1 catalyzes NAD-dependent protein deacetylation and activates PGC-1α, an essential cofactor in mitochondrial biogenesis that controls glucose metabolism and improves skeletal muscle metabolism.[6] These effects reduce hyperglycemia and hyperlipidemia, and help to prevent brain diseases such as dementia.[7] It is also reported to modulate higher-order brain functions such as synaptic plasticity and memory formation.[8,9]

Recently, agents controlling the activation of SIRT1 have been identified among natural compounds. Resveratrol, which is abundant in grape seeds, reduces high-fat diet-induced weight gain and increases mitochondrial energy metabolism in association with SIRT1 activation.[10] In addition to resveratrol, several other polyphenolic compounds activate and elevate SIRT1, including butein, piceatannol, fisetin, and quercetin.[11] However, the short half-life and low bioavailability of these compounds limit their clinical application.[12]

Woohwangcheongsimwon (WHC; known as ''Niu Huang Win Xing Wan'' in China) is a widely prescribed traditional oriental medicine, which has been used to treat cerebrovascular diseases such as stroke, delirium, and convulsions.[13,14] Based on these traditionally known effects, WHC may be a drug with tranquilizing or sympatholytic effects.[15,16] But recent experimental research are revealing more prolonged effects of WHC suggesting that it may prevent hypoxic and

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stress-associated brain cell damage, suppress neuroinflammation, and improve long-term memory in animal models.[17–19] As a mixture of various natural products, WHC is expected to have numerous effects, but few have thus far been scientifically confirmed. However, recently interest has been paid to its efficacy as an intervention for metabolic disorders.

On the basis of these results, we investigated whether WHC can prevent cognitive decline in high-fat diet-fed animals. In addition, we investigated whether WHC has SIRT1-inducing activity.

MATERIALS AND METHODS

Reagents

All reagents were purchased from Sigma-Aldrich (St. Louis, MO, USA) unless noted otherwise.

Animals

Ten-month-old male ICR mice purchased from Samtaco (Suwon, Korea) were housed under constant temperature and humidity with 12-h cycles of light/darkness. Animal studies were conducted in accordance with the Guide for the Care and Use of Laboratory Animals by National Institutes of Health. The protocols were approved by the Institutional Animal Care and Use Committee of Kyung Hee University (KHMC-IACUC: 10-071).

Preparation of WHC

WHC was purchased from Kyung Hee Hanyak, Co. (Seoul, Korea). A voucher specimen was deposited in the Herbarium of Kyung Hee University, College of Oriental Medicine. The composition of a daily dose (2.77 g) of WHC is listed in Supplementary Table S1 (Supplementary Data are available online at www.liebertpub.com/jmf).

Treatment

Thirty-six mice were divided into four groups: (1) DW (distilled water) and ND (normal diet) group (DW+ND), (2) DW and HFD group (DW+HFD), (3) WHC and ND group (W+ND), and (4) WHC and HFD group (W+HFD). For a period of 8 weeks, mice in the ND groups were provided with AIN-76A No. 100000 (Dyets, Inc., Bethlehem, PA, USA; Supplementary Table S2). The HFD groups were provided with AIN-76A + 40% beef Tallow No. 101556 (Dyets, Inc.; Supplementary Table S2). For the 8-week period, 40 mg/kg of WHC was administered once a day via oral gavage. During the experimental period, body weight and feed consumption were measured once a week.

Passive avoidance test

The test apparatus consisted of bright and dark compartments (Jeung-Do Bio & Plant, Co., Ltd., Seoul, Korea) with the mice being allowed to acclimatize to it for 2 min. As a pretest, the mice were placed into the bright compartment and the latency to stepping into the dark compartment was recorded. For avoidance training, mice were placed into the bright compartment and when they moved into the dark compartment, 0.45-mA electric shocks were administered twice. After 24 h, each mouse was returned to the bright compartment and the latency to entering the dark compartment was recorded to serve as a measure of long-term memory.

Y-maze

Mouse behavior with regard to the arm(s) visited in a Y-shaped maze was videotaped for 5 min to provide a measure of alternation behavior reflecting short-term memory. One alternation was defined as successive consecutive entries into the three different arms on overlapping triads in which all arms were represented. Percent alternation was calculated using the following equation;

$$(\text{total alternations}/[\text{total arm visits} - 2]).$$

Immunoblotting

Mice were sacrificed by decapitation after an overdose of ether anesthesia and their brains were extracted. The hippocampus was separated and homogenized in lysis buffer[20] by repetitive suction and subsequent sonication. The homogenate was resolved by SDS-PAGE (sodium dodecyl sulfate-polyacrylamide gel electrophoresis) and transferred to a PVDF (polyvinylidene fluoride) membrane (Thermo Fisher Scientific, Waltham, MA, USA). The membrane was blocked for 1 h and anti-SIRT1 or anti-PGC-1α antibodies (Santa Cruz, CA, USA; 1:1000) were reacted with the membrane at 4°C overnight. The membrane was then incubated with HRP (horseradish peroxidase)-conjugated secondary antibody (1:1000; Jackson ImmunoResearch Lab, Inc., West Grove, PA, USA) for an hour. An enhanced chemiluminescence solution (Thermo Fisher Scientific) was applied for 5 min, and the protein band was then visualized with a Davinch-Chemi imaging system (CellTAGen, Seoul, Korea). Quantitative analysis of the band density was performed using ImageJ (NIH, USA).

High-performance liquid chromatography

A Sunfire C18 Column (250 × 4.6 mm, 5 μm; Waters Corporation, Milford, MA, USA) was used with a Waters System 600 controller and 996 PDA detector. WHC (100 mg) was melted in 10 mL of 85% MeOH and sonicated for 15 min. For the high-performance liquid chromatography (HPLC) analysis, 5–70% acetonitrile (A) and 1% phosphoric acid (B) were used. The sample analysis time was 0–60 min. The solvent speed was maintained at 1 mL/min, with 10 μL of sample being injected.

Cell culture and reverse transcriptase polymerase chain reaction

HT22 cells cultured in Dulbecco's modified Eagle's medium with 10% added fetal bovine serum were treated with either vehicle, resveratrol 2 μM, or WHC 0.5–100 μg/mL. After 2 days, cells were harvested and homogenized with TRIzol for messenger RNA (mRNA) and protein extraction. SIRT1 reverse transcriptase polymerase chain reaction (RT-PCR) were performed with forward 5′-CAGTGTCATGGTTC CTTTGC-3′ and reverse 5′-CACCGAGGAACTACCTGAT-3′ primers for 35 cycles, and β-actin RT-PCRs were performed with forward 5′-GCTCCGGCATGTGCAA-3′ and reverse 5′-AGGATCTTCATGAGGTAGT-3′ primers for 25 cycles.

Statistics

An analysis of variance and Tukey's honest significant difference test were performed with Sigma Stat 3.5 (Systat Software, Inc.). A P value <0.05 was considered statistically significant.

RESULTS

WHC reduced weight gain and memory deficits in high-fat diet-treated mice

The body weight of high-fat diet-fed mice increased during the latter half of the experimental period (Fig. 1A). WHC had no significant influence on body weight in the ND fed mice, but reduced weight gain caused by the HFD. The prevention of body weight gain was not due to appetite change, as the daily energy intake was not changed by WHC (Fig. 1B). After the HFD, the plasma concentrations of high-density lipoprotein (HDL) and low-density lipoprotein (LDL)

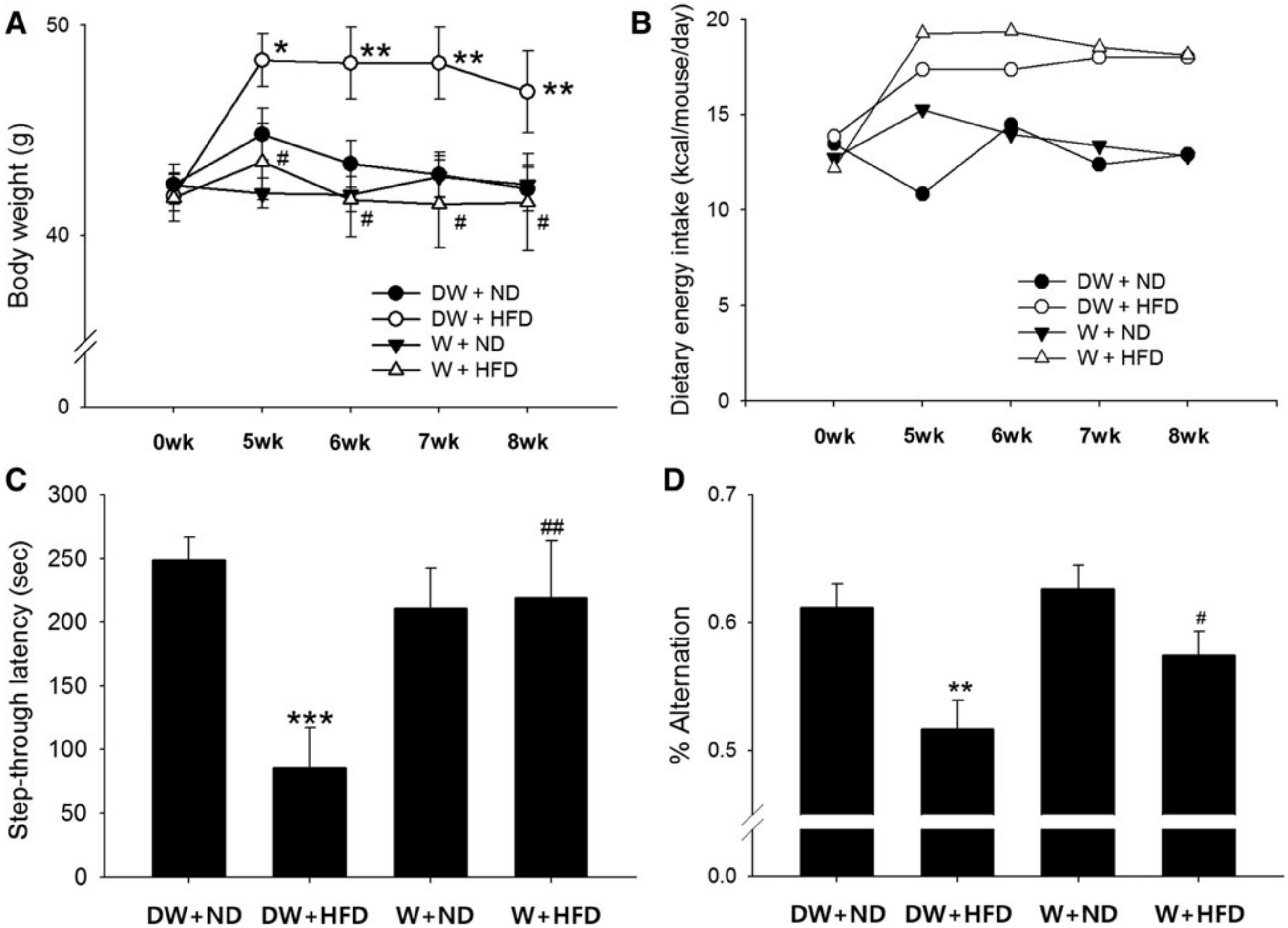

FIG. 1. The effects of WHC on body weight, caloric intake, and cognitive behavior. **(A)** Body weight change during the experiment. A difference between the groups [$F(3,189) = 11.8$; $P < .0001$] was observed. WHC had no effect on animals under a normal diet (DW+ND vs. W+ND), but it reduced weight gain in animals under a high-fat diet (DW+HFD vs. W+HFD). **(B)** Dietary energy intake during the experiment. The average daily caloric intake per mouse differed between groups [$F(3,19) = 7.3$, $P = .003$]. A pairwise comparison revealed an increased caloric consumption in DW+HFD ($P = .019$) and W+HFD ($P = .007$) groups in comparison with the DW+ND group. Caloric consumption was also higher in DW+HFD ($P = .07$) and W+HFD ($P = .028$) groups in comparison with the W+ND group. **(C)** Step-through latency 24 h after avoidance training. An inter-group difference existed [$F(3,36) = 5.78$; $P = .0025$]. Long-term memory defects were found in high-fat diet-fed animals, as measured by the shortening of latency (DW+ND vs. DW+HFD; $P < .001$). WHC itself had no improvement-related properties in normal mice (DW+ND vs. W+ND; $P = .31$), but it prevented the deteriorating effect of the high-fat diet (DW+HFD vs. W+HFD; $P = .007$). **(D)** Percent alternation in a Y-maze. A difference in short-term memory was noted between the groups [$F(3,32) = 6.4$; $P = .016$]. The high-fat diet reduced percentage alternation (DW+ND vs. DW+HFD; $P = .0015$), but WHC administered with a high-fat diet improved the percentage alternation (DW+HFD vs. W+HFD; $P = .045$). $^{*}P < .05$, $^{**}P < .01$, $^{***}P < .001$ vs. DW+ND. $^{\#}P < .05$, $^{\#\#}P < .01$ vs. ND+HFD. Two-way ANOVA for **(A)**, ANOVA for **(C)** and **(D)**, Tukey's HSD *post hoc* test. ANOVA, analysis of variance; DW+HFD, distilled water and high-fat diet; DW+ND, distilled water and normal diet; HSD, honest significant difference; W+HFD, WHC and high-fat diet; W+ND, WHC and normal diet; WHC, Woohwangcheongsimwon.

increased, although concentrations of triglycerides (TG) did not; however, WHC had no effect on the lipid contents (Supplementary Fig. S1).

While there were no differences between the experimental groups in the pretest latency to move into the dark compartment, the long-term memory performance in the passive avoidance paradigm showed that the mice fed the HFD had shorter step-through latencies than the ND group, whereas the long-term memory of the mice that were also fed WHC was improved (Fig. 1C). However, there was no prolongation of latency in the normal mice treated with WHC. This indicates that HFD disrupts avoidance memory, and that the addition of WHC helped to prevent this disruption.

As a measure reflecting short-term memory, the percent alternation into each arm of the Y-maze decreased in the HFD mice (Fig. 1D). Addition of WHC did not change the performance in the Y-maze in the ND animals, but did improve the performance of HFD animals. However, differences in ambulatory activity did not confound this finding, because total arm entry numbers did not differ among the experimental groups. According to these results, WHC prevented high-fat diet-induced weight gain and memory deficits.

WHC induced the expression of SIRT1 and PGC-1α

The HFD and WHC had a significant effect on hippocampal expression of SIRT1 and PGC-1α (Fig. 2). Immunoblotting and densitometry revealed reductions in both SIRT1 and PGC-1α in the HFD mice. On WHC treatment, hippocampal expression of SIRT1 and PGC-1α increased in both the ND and HFD groups (Fig. 2B, C). These results demonstrate that WHC induced the SIRT1 protein and the downstream effector PGC-1α.

Resveratrol was not detected in WHC

A few polyphenolic compounds have demonstrated SIRT1-activating properties, including resveratrol.[11] HPLC was performed to determine whether resveratrol was a component of WHC, which might explain its SIRT1-inducing activity (Fig. 3). Analysis revealed that WHC did not contain resveratrol. The compounds glycyrrhizin (Supplementary Fig. S2) and baicalin (Supplementary Fig. S3) were found in WHC by chromatographic analysis, but betulinic acid (Supplementary Fig. S4) and allantoin (Supplementary Fig. S5) were not.

WHC induced in vitro *SIRT1 expression*

SIRT1 mRNA and protein expression in HT22 cells were measured after 48 h of WHC or resveratrol treatment. SIRT1 mRNA expression was increased by 2 μM resveratrol ($P < .01$), and 0.5 and 1 μg/mL WHC ($P < .05$; Fig. 4A). SIRT1 protein expression was also increased by 2 μM resveratrol ($P < .01$), and 0.5 ($P < .01$) and 1 μg/mL WHC ($P < .05$; Fig. 4B). These results indicate that WHC and resveratrol are possible hippocampal SIRT1 inducers.

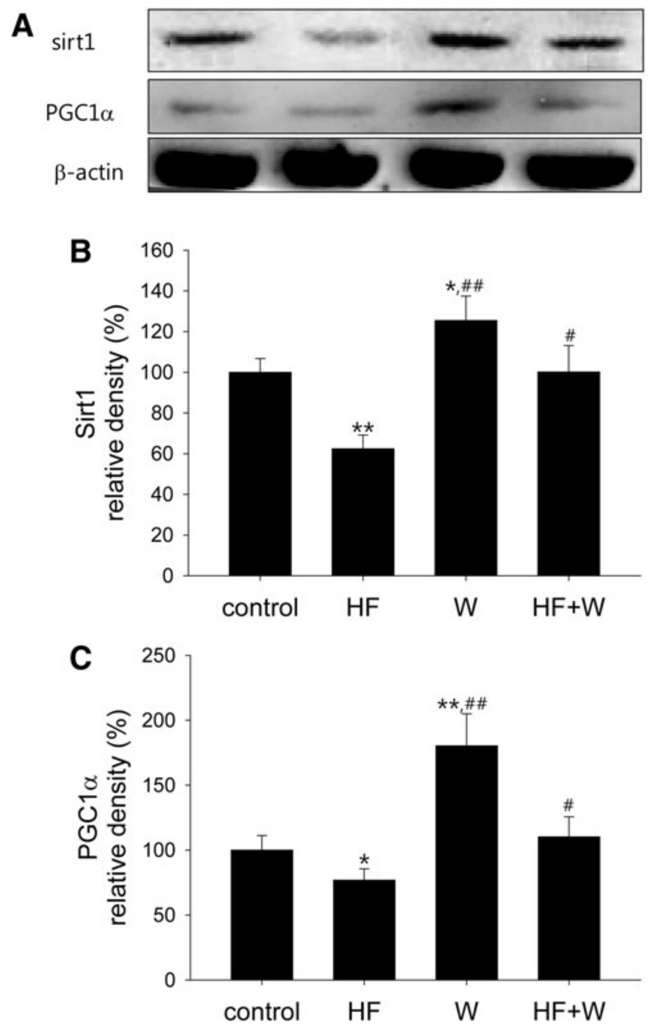

FIG. 2. The effects of WHC and a high-fat diet on hippocampal expression of SIRT1 and PGC-1α. **(A)** Representative blotting image of hippocampal SIRT1 and PGC-1α expression. **(B)** Differences existed with regard to SIRT1 [$F(3,33) = 6.24$, $P = .0018$] expression. SIRT1 expression decreased in the presence of a high-fat diet (DW+ND vs. DW+HFD; $P = .008$), increased after treatment with WHC (DW+ND vs. W+ND; $P = .047$), and increased after treatment with WHC in high-fat diet mice (DW+HFD vs. W+HFD; $P = .021$). **(C)** Differences existed in hippocampal PGC-1α expression [$F(3,33) = 5.76$, $P = .012$]. PGC-1α expression was decreased by a high-fat diet (DW+ND vs. DW+HFD; $P = .03$), increased by WHC (DW+ND vs. W+ND; $P = .004$), and increased by WHC in high-fat diet mice (DW+HFD vs. W+HFD; $P = .036$). $^{*}P < .05$, $^{**}P < .01$ vs. DW+ND. $^{\#}P < .05$, $^{\#\#}P < .01$ vs. DW+HFD. ANOVA, Tukey's HSD *post hoc* test.

DISCUSSION

A HFD is known to trigger metabolic deterioration, and to also be a factor in reduced cognitive functions.[21] We report here that WHC reduced weight gain and cognitive deterioration in mice fed high-fat diets, possibly through a link with SIRT1 induction.

High-fat diet-fed mice showed increased body weights, increased plasma concentrations of HDL and LDL, and deficits in avoidance memory and working memory. Previously, a continuous HFD was reported to cause cognitive decline in

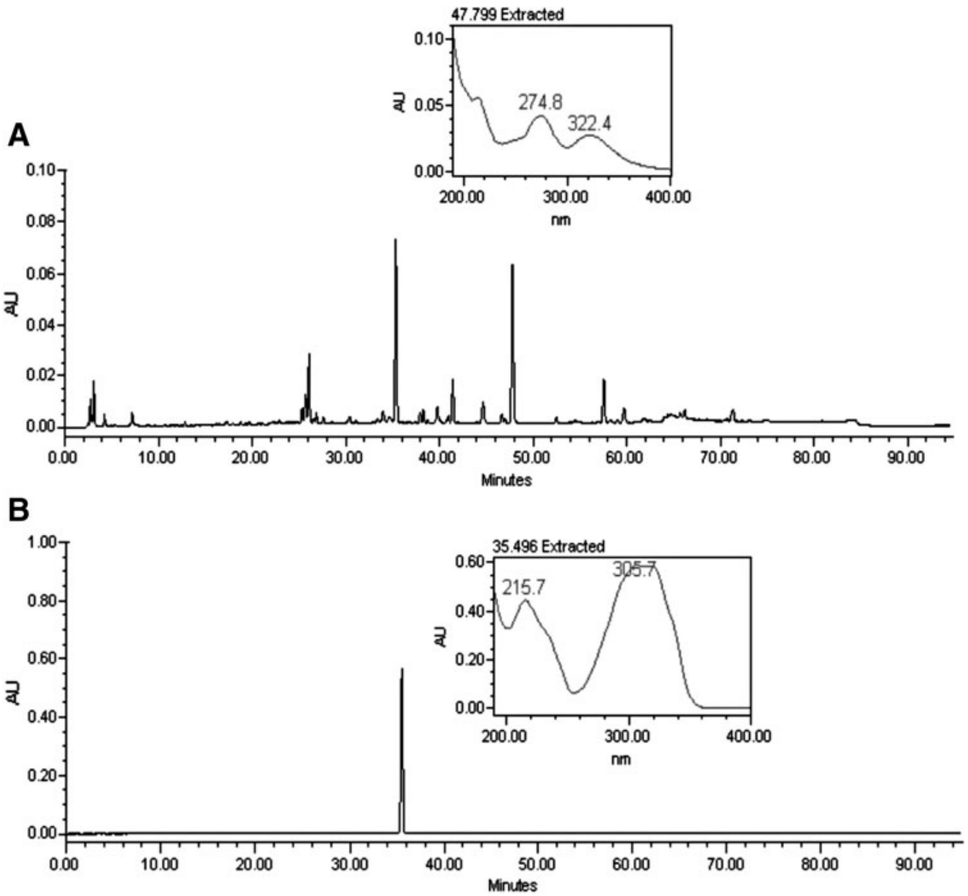

FIG. 3. High-performance liquid chromatograms of WHC and resveratrol. Chromatographic analyses of WHC **(A)** and resveratrol **(B)** showed similar peaks at 36 min. However, spectral analyses produced different patterns for WHC than for resveratrol. This confirmed that resveratrol was not a component of WHC.

rodents, accompanied by inflammatory changes in the hippocampus.[2] HFDs also accelerate amyloid β deposition in Alzheimer's disease animal models.[22] No significant effect was observed when WHC was administered to normal mice, but in mice fed a HFD body weights remained at normal levels, and avoidance memory and working memory impairments due to HFDs were not found. However, WHC did not reduce the elevation of HDL and LDL by the HFD.

Despite the high calorie intake in the two groups that received HFDs, WHC was associated with lower weight

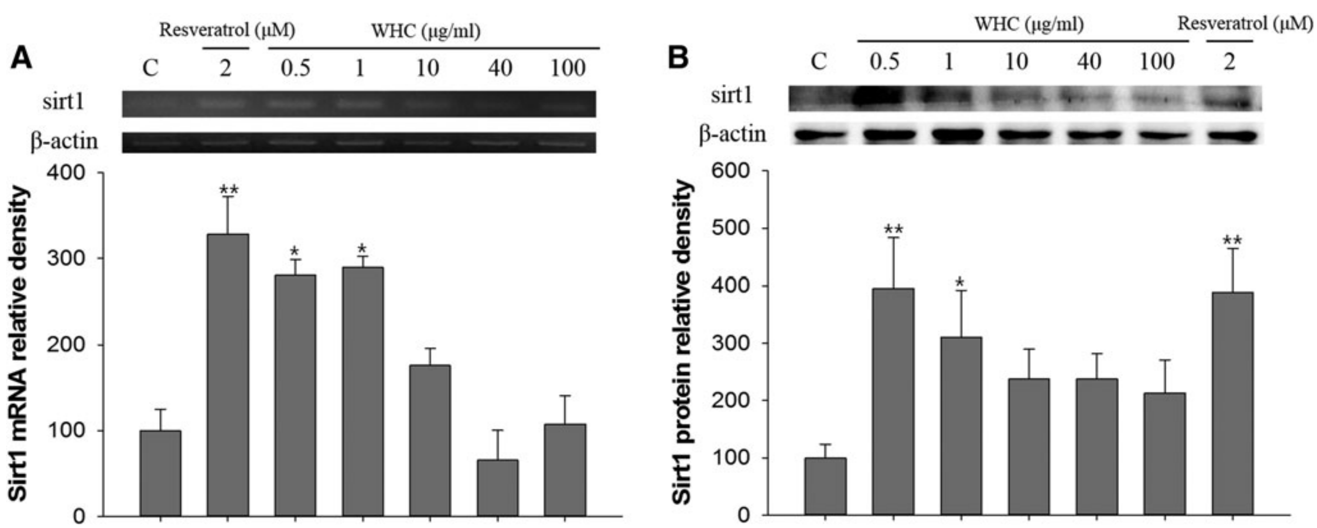

FIG. 4. SIRT1 mRNA and protein induction by WHC and resveratrol in hippocampal origin HT22 cells. **(A)** RT-PCR ($n = 3$) **(B)** Western blot ($n = 3$). $^{**}P < .01$, $^{*}P < .05$ vs. control. ANOVA, Tukey's HSD *post hoc* test. C, vehicle-treated control; mRNA, messenger RNA; RT-PCR, reverse transcriptase polymerase chain reaction.

gain, which is presumed to be the result of WHC increasing caloric expenditure. With regard to the memory improvement, WHC has been shown to improve referential long-term learning in nitric oxide synthase inhibitor-treated mice.[19] It also reduced neuroinflammation by suppressing TNF-α, IL-1β, iNOS, and COX-2, and activating microglia,[18] and reduced stress-associated oxidative damage.[17] On the basis of these results, it is likely that the WHC-induced cognitive function improvement in mice fed on a HFD is associated with the inhibition of inflammatory mediators and anti-inflammatory effects.

Next, we examined the role of SIRT1 in relation to the increase in caloric expenditure in mice fed on HFDs. SIRT1 is activated by many polyphenols and can reduce weight gain through facilitated caloric usage.[23] Metabolically, SIRT1 stimulates thermogenesis in adipose tissues and can suppress transcription factors involved in lipid synthesis.[24] SIRT1 was reduced in the hippocampi of animals with memory defects after HFD feeding,[25] while weight gain due to aging was prevented by SIRT1 through the improvement of leptin sensitivity.[26] Furthermore, dietary supplements with SIRT1-activating molecules have been shown to reduce weight gain in humans and animals.[27,28] Cognitively, when SIRT1 protein was overexpressed in dementia-model mice, learning and memory improved,[8] while memory and LTP production were impaired when SIRT1 was suppressed.[9] Therefore, these findings and the belief in life-prolonging effects have led to studies of agents that can activate SIRT1.

WHC increased SIRT1 expression in the hippocampus of normal mice, and also increased the SIRT1 expression levels, which was decreased by a HFD. WHC also increased the expression of PGC-1α, a downstream target of SIRT1, and prevented the decrease of PGC-1α by HFD. The memory deficit and decreased expression of SIRT1 protein in the hippocampus by HFD are consistent with reports showing that a decrease in hippocampal SIRT1 expression increases the incidence of hyperlipidemia-induced dementia.[25] The fact that WHC prevented a decrease in SIRT1 expression indicates the possible efficacy of WHC as a treatment for hyperlipidemia-related memory deficit. PGC-1α is a transcriptional coactivator of genes regulating energy metabolism and mitochondrial biogenesis, and is suggested as a therapeutic target for obesity or diabetes.[29] PGC-1α regulates mitochondrial biogenesis and promotes caloric usage through the alteration of the thermogenic program.[29] Additionally, a prolonged calorie-restricted diet resulted in upregulated PGC-1α in the hippocampus.[30] Increased expression of PGC-1α by WHC may have induced a change in energy metabolism, and it may have reduced the side effects of the HFD.

WHC is a formulated decoction of natural herbs and contains a large number of compounds (Supplementary Table S1). From this mixture of compounds, we tried to identify resveratrol,[31] but after an elaborate analysis, we ruled out the possibility of its presence in WHC. Instead, WHC may contain other known sirtuin-activating compounds, such as butein, piceatannol, fisetin, and quercetin, or it may contain an entirely new sirtuin activator.

We confirmed the increased expression of SIRT1 by WHC in the hippocampal HT22 cell line. After 2 days of WHC treatment, SIRT1 mRNA and protein levels were increased at lower doses of WHC (0.5 and 1 μg/mL). Interestingly, the expression of SIRT1 was also increased by resveratrol, which is known to be activator of SIRT1.

The above results led to the following conclusions: (1) WHC increases expression of SIRT1, (2) WHC can prevent memory deficit induced by a HFD, (3) increases in SIRT1 and PGC-1α protein expression in the hippocampus may have reduced memory deficits, and (4) resveratrol was not found in WHC. We suggest that the induction of SIRT1 and PGC-1α expression by WHC promoted energy usage, decreased weight gain, and prevented memory deficits. These cognitive and metabolic benefits are in addition to its traditional use in cardiovascular and neurological emergencies, although there is a translational gap between animal study and human application, and future studies are needed to support this idea. WHC may have antiobesity benefits and prevent memory impairments in excess calorie-consuming individuals.

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AUTHOR DISCLOSURE STATEMENT

No competing financial interests exist.

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